

Sai Raghu Teja Davuluri

[Email](#) | [LinkedIn](#) | [GitHub](#) | San Francisco, CA

Data scientist with 3.5+ years of experience in Market Analytics and research in Computer Vision, CNNs and NLP.

Education

Master of Science in Data Science - *University of San Francisco, San Francisco, CA*

Jun 2023 – Jun 2024

Relevant Coursework: Machine Learning, Data Acquisition, Statistics, Data Analysis, Relational Databases (SQL), Data Ethics, Time Series, Computer Science, Spark, ML Ops, NLP, Airflow, MongoDB

Bachelor of Technology in Electronics and Communication - *Manipal Institute of Technology, India*

Jul 2017 – Aug 2021

Professional Experience

Machine Learning Engineer

Oct 2023 – Present

Isazi

San Francisco, CA, USA

- Conducted intricate data analysis and integrated 35k+ grocery images to label and generate train data resembling in-app pages which are unique to specific products.
- Deployed a fine-tuned Pix2Struct transformer model, achieving a 5% trade-off error to optimize image capture and analysis in Isazi's retail shelf analytics app, ensuring real-time precision in shelf, price, and product identification on mobile devices.

Data Scientist

Jul 2021 – Jun 2023

Merkle

Bangalore, KA, India

- Conducted A/B testing and implemented customer segmentation strategies based on ML clustering techniques, resulting in a 20% uplift in click-to-open rate
- Built incremental propensity models using Xg-Boost to increase ROAS by 40%
- Scored the US population monthly on propensity models in R and SAS and prepared the target data by demographic and behavioral analysis relevant to each campaign and optimized each customer interaction to drive future sales and brand loyalty
- Performed SQL operations on snowflake to obtain the data and leverage mainstream analytical products for regular data analysis for market campaigns

Intern Analyst

Mar 2021 – Jul 2021

Merkle

Bangalore, KA, India

HR attrition project

- Leveraged advanced statistical techniques and implementing a Logistic-XGBoost ensemble model to predict attrition among a dataset of 30,000 employees.
- Generated key insights and KPIs aimed at reducing the hiring rate by 15%, focusing on factors such as employee satisfaction, career development opportunities, and managerial effectiveness.

Uplift Propensity Modeling

- Developed a predictive model using Uplift methodology to identify individuals responsive to targeted marketing emails, incorporating a stacking classifier algorithm with logistic regression, random forest, and XGBoost on a data of 50M records.
- Conducted extensive exploratory data analysis and feature engineering to optimize model performance, achieving an AUC of 74% while neutralizing the effects of underfitting and overfitting.

Co-Founder

Apr 2020 – Dec 2020

Pergo.io

Rajahmundry, AP, India

- Co-founded an e-commerce startup, overseeing product development, marketing, and achieving INR 80 lakhs in revenue.
- Directed the team to deliver high-quality products, driving customer satisfaction and earning 4.3/5 rating on the Play Store.

AI Researcher

Sep 2017 – Aug 2019

Mars Rover Manipal, Student Project, MIT

Manipal

- Designed autonomous traversal algorithms and object-tracking models for rover navigation
 - Coded to map and acquire data from soil and biosensors to give high-precision
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Select Research Projects

Lane detection algorithm using auto-encoder convolutional architecture (CV, CNN, LSTM) – [GitHub](#)

- Built a deep convolutional encoder-decoder architecture called SegNet integrated with LSTM architecture to detect the drivable area. The second part of the algorithm consists of a series of operations, including Canny Edge Detection, Perspective Transform, Sliding Window, and Polynomial Curve Fit.
 - Achieved a test accuracy of 93.5% and presented at the **IEEE conference — Student Branch Manipal**.
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Achievements

- Secured eighth position globally in University Rover Challenge 19 held at Mars Desert Research Society (MDRS), Utah
 - Awarded AICTE scholarship of 8L INR by the government of India for four years of my undergraduate course.
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Technical Skills

Python, R, SQL, Postgresql, Linux, GPT, BLIP, LLMs, HTML, AWS, GCP, Tableau, Snowflake, Pytorch, Scipy, Tensorflow, OpenCV, DataBricks.